



Asset Performance: Ene 06 Heating



Contributes to 40 credits
available for asset energy
rating

No Minimum Standard

Aim

To minimise operational energy consumption and the associated carbon emissions by increasing the intrinsic energy efficiency of the building fabric and the installed buildings services.

Question

1. Is space heating generated on-site?
2. What is the main generation type for on-site space heating?
3. Please enter the efficiency or COP of the main on-site heat generator.
4. What is the main fuel used for on-site heat generation?
5. Is the on-site main space heating system centralised or local?
6. Is heat distributed around the building via air?
7. In what year was the main heat generator/heating system last replaced (if known)?
8. For district heating only:
 - a. Do you know what types of heat generation are used for the network heat supply?
 - b. What are the main heat generation sources?
 - c. What is the percentage distribution losses for the heat network (if known)?

Credits	Question	For questions 1 - 6, 8 a - b select a single answer. For questions 7 and 8c enter value
-	1.	Yes / No
-	2.	Question not answered Boiler Heat pump/reversible chiller Direct electricity consumption Other on-site heat generation e.g., CHP/Solar thermal Other (user entry)
-	3.	Enter efficiency

Credits	Question	For questions 1 - 6, 8 a - b select a single answer. For questions 7 and 8c enter value
-	4.	Question not answered Electric Burning oil Coal (domestic) Coking coal Diesel Fuel Oil Gas Oil LNG LPG Natural Gas Other Petroleum Gas Refinery Miscellaneous Biodiesel Landfill gas Charcoal Other Biogas Wood/waste wood Renewable heat source District heating
-	5.	Centralised / Local
-	6.	Yes / No
-	7.	Enter year
-	8.a	Yes / No
-	8.b	Heat source 1 Heat source 2 Heat source 3 Heat source n
-	8.c	%

Methodology

Main generation type for space heating

1. Where there is more than one heat generation type in the asset, the Assessor must verify that the selected generation type is the main heat source for the asset.
2. This should be based upon respective system heat output capacities.

Multiple heat generation units for the main heating system type

If there is more than one heat generation unit the Assessor should calculate the average energy efficiency, or COP for heat pumps of the units based on their respective heat generation capacities.

COP of heat pumps

The coefficient of performance (COP) of a heat pump is calculated as the ratio of heating or cooling provided to electrical energy consumed. COP entered must not be higher than 7.

Efficiency of CHP unit

The efficiency of a CHP unit should be calculated as follows:

$$\text{Overall efficiency} = \text{thermal efficiency} + (2 \times \text{electrical efficiency})$$

As an example, if the thermal efficiency was 50% and the electrical efficiency 35% the calculated efficiency would be:

$$50\% + (2 \times 35\%) = 120\%$$

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Visual inspection and verification through photographic evidence of listed space heating system, indicating the distribution method.
All	Extract of O&M manuals or a copy of manufacturer information for heating systems that are present in the asset.
All	Evidence demonstrating the main heating system is local/ centralised.
All	Installation diagrams of the space heating system.
6	Copy of documentation outlining when the heating system was installed or replaced, such as: a) Extract of O&M manuals or a copy of manufacturer information for heating generator/heating system b) Service records c) Installation records d) Maintenance records
All	Evidence demonstrating when the main heat generator/heating system last replaced
8	Copy of documentation provided by the district heating operator.

Definitions

COP:

The coefficient of performance (COP) of a heat pump is the ratio of heating or cooling provided to electrical energy consumed.

Additional information

Asset energy calculator

In order to obtain the maximum possible number of credits for the asset energy performance as much information as possible should be entered. Where no value is entered, the asset energy rating calculator will assume a pessimistic default value.